
Relationship between crimes and socio-economic factors in Los Angeles

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Abstract

In recent times there has been a motive among researchers to study the issues related to different types of crime and socioeconomic factors across the United States and to understand the relationship between the several underlying intricacies. Over the past decade, the crime rate has increased by over 30% in the greater Los Angeles area and it continues to increase at an unprecedented rate. It is paramount to identify the underlying causes and preconditions of crime which plays a crucial role in the process of crime reduction and prevention. With the advancement of technology in recent years, getting access to crime and census data has become relatively less challenging and it paves the way to predict population-based data at a microscopic level. The sole aim of this research is to identify the correlation between several socio-economic factors (Ex: Unemployment, poverty, educational background, mean income, etc.) of the population and corresponding crime occurrences in the city of Los Angeles. In addition to this, the experiment also aims to identify the hotspots of crime and evaluate the major contributing factors to such crimes.

1 Introduction

On any typical day, we come across different types of news through several forms of media. Crime reports are a major part of today's events and one can always concur that crime reports bring a sense of shock to the minds of the people. The level of crime has immensely skyrocketed over the years across the United States, and several research results reveal that the factors which influence the crime rate in our society are including but not limited to drug/alcohol abuse, unemployment, poverty, politics, hate crime, economic background, and religion. The crimes mentioned above can be segregated into categories like theft, murder, sexual violence, assault etc. and each type of crime may have different reasons or factors associated with it.

1.1 Problem Statement

The problem we are trying to address is to find a correlation between the types of crimes and the socio-economic factors that influence such crimes. To understand this correlation in detail and to provide a more concentrated solution, it is pivotal to evaluate crime rates and recognize the preconditional patterns. This research will focus on the city of Los Angeles and its peripheries to produce a factual representation that displays the relationship between socioeconomic factors and the crime data.

1.2 High Level Goals and Approaches

Previous research indicates that low socioeconomic status is related to a higher crime rate. We want to explore deeper into this fact and present our findings in an elaborate manner. Below is the list of goals and approaches that this project majorly focuses on.

- With the information collected from various sources like census data, we initially collated various datasets including Crime, Population Density, and Household income for the city/county of Los Angeles.
- The next step focused on aggregating the collected data units to similar geographic levels and cleaning the data by eliminating unwanted variables to maintain data consistency.
- With the use of regression models for feature selection we were able to optimally determine the set of features that has maximum correlation magnitude to the crime dataset. After this, we will use multiple regression models on future data like mean household income, rental prices, and population estimates gathered from different sources and predict the future crime rate for the city of LA.
- Overall, we aim to identify features representing socio-economic patterns in violent crimes committed over the years as well as to predict violent crime rate for future years.

The problem we are trying to solve matches well with the SDGs like #3 Good Health and well-being and #11 Sustainable Cities and communities.

2 Background and Related work

There are multiple research projects done in this space and some are mentioned below. A research study by the American psychological association suggests that socioeconomic disadvantage (usually measured using the three variables income or wealth, occupational level, and years of education) is associated with clear increases in rates of both self-reported crimes and officially recorded convictions. In addition, the Economic theory also predicts a positive correlation between different types of crime and their underlying economic factors. (Becker, 1968; Freeman, 1996). More specifically, Arvanites and Defina used a more localized economic indicator – Gross State Product (GSP) to measure economic health and found a significant relationship between GSP and the crime rate from 1986 through 2000.

Researchers often argue that crime is influenced by more prominent indicators like the unemployment rate and political intricacies. However, research by Rosenfeld and Fornango that used annualized values from the Index of Consumer Sentiment (ICS) indicated that social factors were more highly correlated with robbery and other crimes than more traditional measures of the economy, such as the unemployment rate.

Research by Marco De Nadai, Yanyan Xu shows that the combined use of socio-economic conditions, mobility information, and physical characteristics of the neighborhood effectively explain the emergence of crime and improve the performance of the traditional approaches in predicting the relationship between different types of crime and its underlying socioeconomic factors.

2.1 Improvements from Previous work

The previous research done in this field utilizes the crime and socio economic data comprised from a much smaller timeframe. One other thing to be noted is that the prior research done in the similar field performs manual feature selection from a relatively smaller set of features while our research aims to statistically and strategically select features that have a larger magnitude of correlation from a larger feature set.

3 Data

This section provides an overview of LAPD crime data along with the American Community Survey (ACS) 5-year estimates data. The data was downloaded using Census Bureau's API and LAPD database and was merged on a census tract level.

3.1 Data Description

3.1.1 LAPD Crime Data:

This dataset reflects incidents of crime in the City of Los Angeles from 2010 - 2019. The dataset consists of 2 million rows and 28 columns of crime data including crime codes, area, date of incident, latitude, and longitude details.

3.1.2 American Community Survey (ACS) 5-year Data (2009 - 2020):

The American Community Survey is the premier source for information about America's changing population, housing and workforce. The data from this survey covers a broad range of topics about social, economic, housing, and demographic information on the U.S. Population. The dataset consists of around 28k rows and 4k columns. We have used 5-year estimates as they help us in analyzing very small populations with increased statistical reliability of data. This dataset is also helpful for examining census tracts and other smaller geographies because 1-year estimates are not available.

3.2 Data Sources

We used the Census Bureau's Application Programming Interface (API) tool to access American Community Survey (ACS) 5-year estimate data in its entirety from the United States Census Bureau website.

For the crime data, we accessed the downloaded given on the Data Catalog of the City of Los Angeles' Open Data.

3.3 Data Statistics - Source Data

The crime data consists of more than 2 million records with 28 features. More summary statistics regarding the crime dataset are mentioned in the table below.

<i>Crime Codes Range:</i>	100 - 1000
<i>Latitude and Longitude:</i>	68479 unique locations
<i>Crime Code Types:</i>	142 (including violent crimes)
<i>Incident Year Range:</i>	2010 - 2019
<i>Average Crimes per year:</i>	212k

The ACS census tract data consists of 28k records and 4k features of population counts and percentages.

<i>Unique Census Tract:</i>	3177 (LA County tract)
<i>Year Range of Census Tract:</i>	2009 - 2020
<i>State and County:</i>	California - LA County

3.4 Data Pre-processing

Crime Data preparation:

- Geo-coding: The Federal Communication Commission's Census Block Conversions API was used to extract the US Census Bureau Census Block number (aka the 15 character FIPS Code) for each data point using its Latitude and Longitude. The census tract numbers were

then extracted from the block number values. This process was performed for each latitude and longitude value in the crime data to geo-code it to the same level as the census data.

- Filtering: Filter crime data for violent crimes (codes up to 300), 13 crime codes categories out of 142 original representing violent crimes.
- Aggregate: The crime data was grouped by year and census tract. Aggregate values were calculated for each of these groups with the total number of crimes reported for each group.

Census Data preparation:

- The ACS data extracted from 4 different source tables is merged into one single file using the census tract and year as unique identifiers.
- Duplicate columns are removed from the dataset.
- Unwanted features containing margin of error and annotation variable details from the data are removed.
- Columns that do not have data till 2016 are also removed.
- Replaced -88888888, NaN, etc with Null and then remove data with more than 33 percent Null Values. (These values represent missing and unavailable data points.)
- Replaced the remaining Null values with 0.

Finally, we merged the Census data and Crime data on a Census tract level for the City of Los Angeles to do our analysis.

3.5 Data Statistics - Processed Data

The processed data, combining the crime data with the census data, consists of 14086 records with 934 features in total. This data maps the year for each of the records with the corresponding Census Tract Information along with the aggregate crime rates.

The data is spread over 1023 unique census tracts for the city of Los Angeles that has at least one crime reported (of any type) from 2009.

3.6 Data Visualization and exploratory analysis

Fig. 1. shows how the average number of crimes for all census tracts in the crime data varies across the years.

Fig. 2. and Fig. 3. plots each data points in our crime data against its latitude and longitude values. The resulting picture shows roughly the outline of Los Angeles county as a whole. This visualization helps us understand how filtering data to consider only violent crime removes data collection bias from the model. The project does make the underlying assumption that all violent crimes are reported equally across the county.

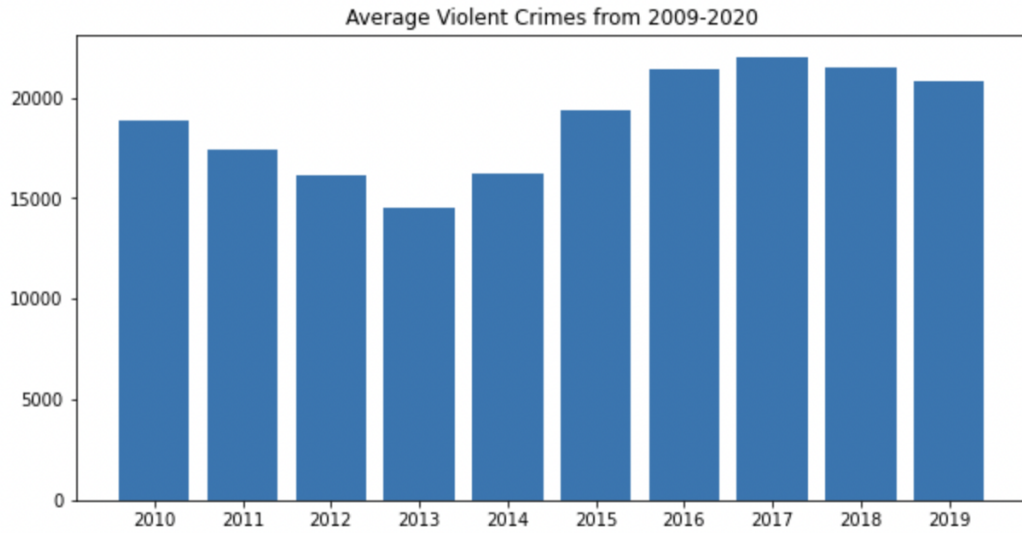


Figure 1: This plot depicts the change in violent crimes reported for the duration of 2010-2019.

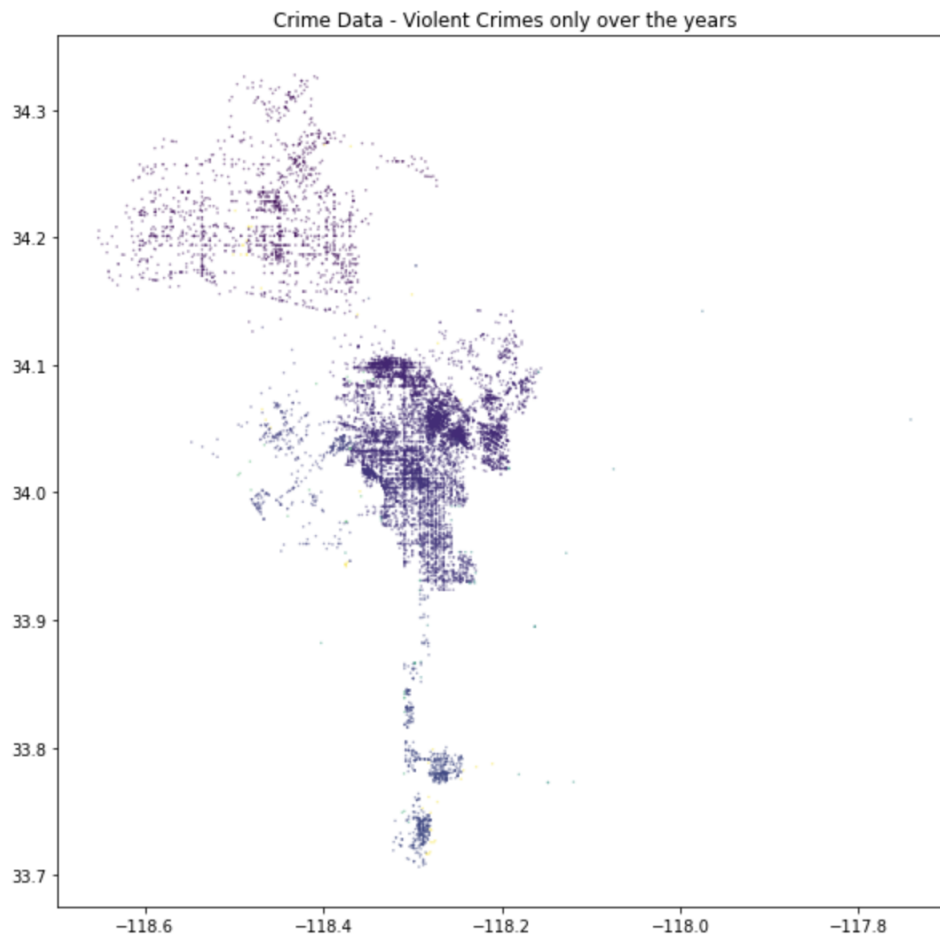


Figure 2: This figure show the spatial location (latitude and longitude) of each data point for the violent crimes. This helps us visualize the frequency of violent crimes reported for each census tract.

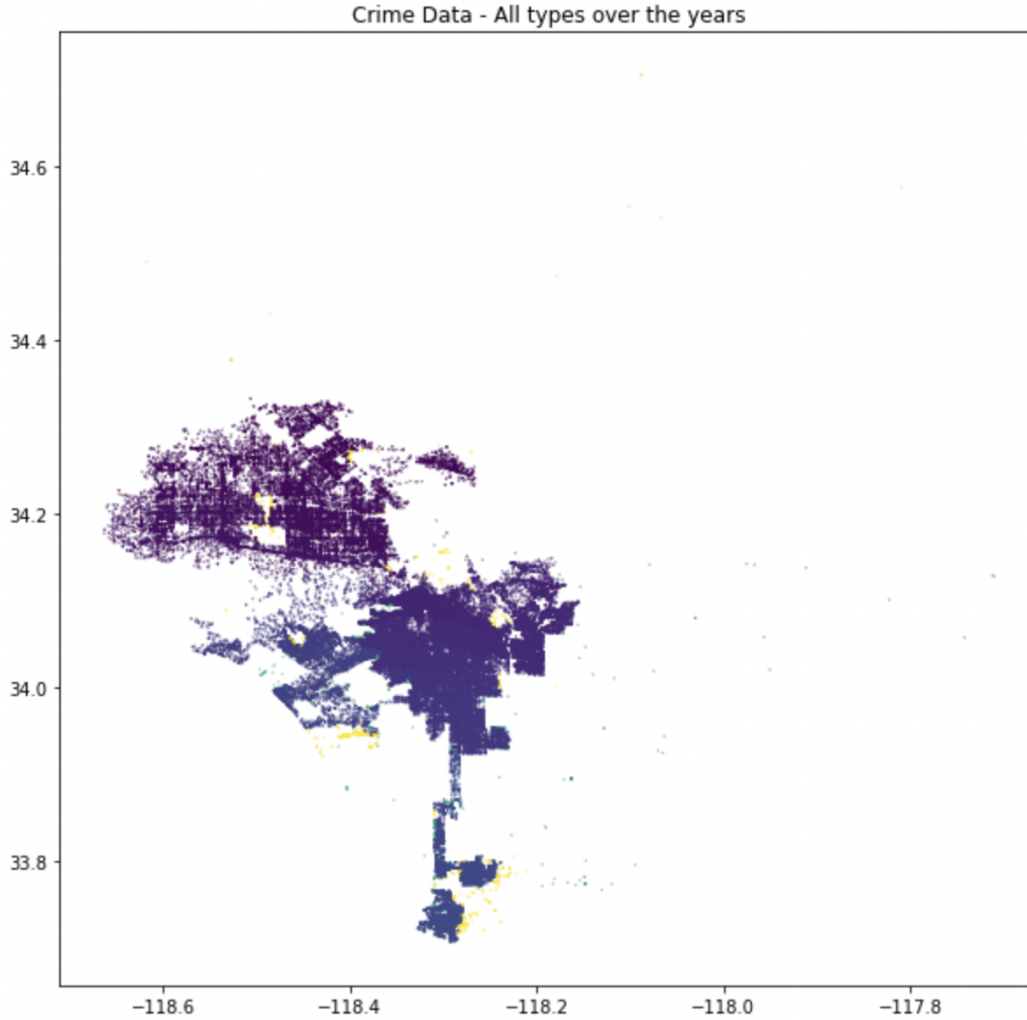


Figure 3: This figure show the spatial location (latitude and longitude) of each of the data points for the all types of crime. This helps us visualize the frequency of all crimes reported for each census tract.

To ensure meaningful results, we performed correlation analysis to find features that are highly correlated. We manually removed 61 highly correlated variables with a similar meaning.

4 Task and Approach

This section covers the details of different tasks and approaches that were implemented for the analysis of the crime and census data. The task at hand is to identify the socioeconomic factors that affect the crime rates for different years and make a prediction for future years. With the data that we have, we aim to identify the most important features from a pool of 945 features and evaluate the model performance. To achieve these results, we tested multiple approaches which are mentioned below.

4.1 Approaches

The data was split into train and test, based on years. The training data consists of data from 2010 to 2018, an inclusive range, whereas the test data consists of data for the year 2019. The training

data consists of 11447 records and the test data has 2639 records spread across all census tracts. This data was used to perform a regression analysis to predict the crime rates for various years by utilizing socioeconomic factors.

4.1.1 Feature selection

In order to identify specific features which contribute significantly to our model performance, we performed Forward Feature Selection (FFS) on Linear Regression Model using multiple target feature numbers like 5, 15, 30, and 45. The features that we got from FFS included the most important socio-economic features that affect the crime rate for each of the years. R-squared value was used as a scoring metric to evaluate the feature importance and we found that the model performance was best explainable when we shortlisted 30 features. With the increase in the number of features, we observed that the model performance did not increase significantly. The results listed in the next section are for the best set of features selected.

4.1.2 Crime Prediction

We tried different regression models for the prediction of the aggregate number of crimes for the future years using the shortlisted features. We implemented Linear Regression, SVM, and Random Forest models, out of which random forest performed the best. SVM's performance was the least, which could possibly be due to linear feature selection.

4.2 Results

4.2.1 Feature selection

Reporting the average R-squared values for cross-validation of feature selection:

<i>5 features</i>	0.407
<i>15 features</i>	0.467
<i>30 features</i>	0.499
<i>45 features</i>	0.513

In feature selection, the increase in accuracy plateaued after a certain point for all models which highlights the need for two things: more correlation analysis as well as non-linear feature selection.

4.2.2 Crime Prediction Accuracy

Reporting the model-wise accuracy metric for each of the models tried on the set of 30 features extracted:

<i>Linear Regression</i>	0.425
<i>SVM Kernel - RBF</i>	0.034
<i>SVM Kernel - Poly</i>	0.029
<i>Random Forest</i>	0.711

The model with 30 selected features shows some interesting results. One such result is the feature listing the population of people paying a rent of \$300-\$399, which is a very low value as compared to Los Angeles City's average rent value which is \$2,734. Another shortlisted feature was the count of people living alone in non-family households. Both of these extracted features align with the feature selection goals of this project.

4.3 Upcoming Work

- Removal of additional correlated variables.
- Implementing more advanced nonlinear techniques for feature selection and using the respective techniques for crime prediction.
- Convert the census tract feature into polygons to capture the spatial property of the data and then implement spatial regression and spatial error analysis.